

RANDOM PROJECTIVE-HYPERPLANE TOGGLING IS A DISJOINT UNION OF CROWN WALKS

ANONYMOUS

ABSTRACT. Let $E = \mathbb{F}_2^d \setminus \{0\}$, and repeatedly toggle the points in the kernel of a uniformly sampled nonzero linear form. We determine this chain on the full 2^{d-1} -state subset carrier. Its communicating classes are 2^{d-d-2} equal cosets, and every class is simple random walk on the crown graph $K_{2^d, 2^d}$ with one perfect matching removed. Fourier inversion gives the exact number of ordered histories between every pair of states at every time. The resulting law converges within its parity-compatible half at exact total-variation distance $1/(2^d(2^d-1)^{t-1})$; it does not mix in ordinary total variation because the chain has period two. On the whole carrier the only eigenvalues are $1, -1, \pm(2^d-1)^{-1}$, with all multiplicities explicit and no Jordan blocks. Simplex-code, incidence-design, Cayley/Fourier, and crown-graph facts are treated as background. The residual claim remains OWNER_AMBER; external circulation is HOLD_EXTERNAL.

1. MODEL, BACKGROUND BOUNDARY, AND THEOREM

Fix $d \geq 2$, put $V = \mathbb{F}_2^d$, $E = V \setminus \{0\}$, and write

$$(1) \quad q = 2^d, \quad N = q - 1, \quad m = |E| = N.$$

We identify a subset of E with its incidence word in $\mathbb{F}_2^E$; addition of words is symmetric difference. At each epoch choose, independently and uniformly, a nonzero form $\ell \in V^* \setminus \{0\}$ and update

$$(2) \quad A \mapsto A + h_\ell, \quad h_\ell = \mathbf{1}_{H_\ell}, \quad H_\ell = \{x \in E : \ell(x) = 0\}.$$

Over $\mathbb{F}_2$, the nonzero forms index the projective hyperplanes without a scalar ambiguity. The zero form is not sampled. In particular, $|H_\ell| = 2^{d-1} - 1$ is odd.

Projective systems and simplex codes are standard coding-theoretic objects [3]; binary codes, symmetric-difference designs, and projective hyperplane sections also have a direct established connection [4]. Walks involving generating sets of Abelian groups provide broader Markov-chain context [2]. Brown's hyperplane-chamber walks [1] are a useful terminology control but have a different state space and update from (2). We assign simplex-code and incidence language, symmetric difference, Cayley-graph reduction, finite-abelian character calculus, the named crown graph, and its elementary spectrum no contribution credit. The calculations below are included to make the literal conjunction self-contained.

For $a \in V^*$ define the evaluation word

$$(3) \quad c_a = (a(x))_{x \in E}, \quad \mathcal{C} = \{c_a : a \in V^*\}, \quad \mathcal{W} = \langle \mathbf{1}, \mathcal{C} \rangle \leq \mathbb{F}_2^E,$$

where $\mathbf{1}$ is the all-one word. Let P be the transition operator of (2) and set

$$(4) \quad K = 2^{m-d-1}.$$

Theorem 1 (complete finite atlas). *The following statements hold for every $d \geq 2$.*

- (i) *The closed communicating classes are precisely the K cosets of $\mathcal{W}$. Each has $2q$ states, all states are recurrent, and the support on each class is the crown graph $K_{q,q}$ with one perfect matching deleted. The chain is simple random walk on that graph and has period two.*
- (ii) *For $t \geq 0$, $A, B \subseteq E$, a t -step history from A to B exists exactly when there is a unique $L \in V^*$ such that*

$$(5) \quad A + B = (t \bmod 2)\mathbf{1} + c_L.$$

When this condition holds, the number of ordered form histories is

$$(6) \quad a_t(L) = \begin{cases} q^{-1}(N^t + N(-1)^t), & L = 0, \\ q^{-1}(N^t - (-1)^t), & L \neq 0, \end{cases} \quad P^t(A, B) = \frac{a_t(L)}{N^t}.$$

Outside (5), $P^t(A, B) = 0$.

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(iii) Put

$$(7) \quad \Phi_t(A) = A + (t \bmod 2)\mathbf{1} + C$$

and let $U_{\Phi_t(A)}$ be uniform on these q states. For $t \geq 1$,

$$(8) \quad \|P^t(A, \cdot) - U_{\Phi_t(A)}\|_{\text{TV}} = \frac{1}{qN^{t-1}}.$$

If π_{A+W} is uniform on the full communicating class, then

$$(9) \quad \|P^t(A, \cdot) - \pi_{A+W}\|_{\text{TV}} = \begin{cases} \frac{1}{2} + \frac{1}{2q}, & t = 1, \\ \frac{1}{2}, & t \geq 2. \end{cases}$$

Thus (8) is parity-phase convergence, not ordinary mixing.

(iv) On the entire 2^m -state carrier, the eigenvalues and algebraic multiplicities are

$$(10) \quad \begin{array}{c|cccc} \lambda & 1 & -1 & 1/N & -1/N \\ \hline \text{mult}(\lambda) & K & K & NK & NK. \end{array}$$

Boolean characters give a complete eigenbasis. Hence there are no other eigenvalues and no non-trivial Jordan blocks.

(v) Within this family, the degree N of one unlabelled component recovers $d = \log_2(N + 1)$. If the total state count is also supplied, then it recovers K as the number of components. This is not a characterization among arbitrary regular bipartite graphs.

For reference, the theorem's quantitative spine is

object	exact value
class decomposition	$K = 2^{m-d-1}$ cosets, each of size $2q$
component support	$K_{q,q}$ minus a matching; degree N ; period 2
endpoint kernel	(5)–(6)
phase convergence	$1/(qN^{t-1})$ in total variation
global spectrum	$1, -1, 1/N, -1/N$ with multiplicities K, K, NK, NK

2. CODE COORDINATES AND CROWN COMPONENTS

Lemma 2. The map $a \mapsto c_a$ is injective and linear, $\mathbf{1} \notin \mathcal{C}$, and $\dim \mathcal{W} = d + 1$. Moreover, the masks h_ℓ , $\ell \neq 0$, generate $\mathcal{W}$.

Proof. Linearity is coordinatewise. If $a \neq 0$, some nonzero x satisfies $a(x) = 1$, so $c_a \neq 0$ and the evaluation map is injective. A nonzero form has a kernel of dimension $d - 1 \geq 1$; hence it cannot equal one on every nonzero vector. Thus $\mathbf{1} \notin \mathcal{C}$, proving $\dim \mathcal{W} = d + 1$.

For every nonzero ℓ and $x \in E$,

$$(11) \quad h_\ell(x) = 1 + \ell(x), \quad h_\ell = \mathbf{1} + c_\ell.$$

Given $u \neq 0$, choose $a \notin \{0, u\}$, which is possible because $d \geq 2$, and put $b = a + u$. Then a, b are nonzero and $h_a + h_b = c_{a+b} = c_u$. Finally $\mathbf{1} = h_u + c_u$, so the masks generate both $\mathcal{C}$ and $\mathbf{1}$. $\square$

Proof of Theorem 1(i). A walk on an additive group remains in the coset of the subgroup generated by its increments. Lemma 2 identifies that subgroup as $\mathcal{W}$, so its cosets are exactly the closed irreducible classes. Their size is $2^{d+1} = 2q$, and (4) gives their number. Finite closed irreducible classes contain only recurrent states.

Fix a coset representative R . The unique code coordinates give a bijection

$$(12) \quad R + \epsilon \mathbf{1} + c_a \longleftrightarrow (\epsilon, a), \quad (\epsilon, a) \in \mathbb{F}_2 \times V^*.$$

By (11), a step indexed by $\ell \neq 0$ sends (ϵ, a) to $(\epsilon + 1, a + \ell)$. It therefore joins $(0, a)$ to every $(1, b)$ except $b = a$, exactly the crown graph $K_{q,q}$ minus its diagonal matching. Every edge changes subset parity because every mask has odd weight. Thus all returns have even length, while repeating any mask gives a two-step return. The period is exactly two. $\square$

3. ORDERED HISTORIES AND THE TWO TV COMPARISONS

After the ordered choices $\ell_1, \dots, \ell_t$, linearity in (11) gives the total increment

$$(13) \quad \sum_{r=1}^t h_{\ell_r} = (t \bmod 2)\mathbf{1} + c_L, \quad L = \sum_{r=1}^t \ell_r.$$

Injectivity of $a \mapsto c_a$ proves the endpoint condition and uniqueness in Theorem 1(ii).

Proof of (6). The additive characters of V^* give the exact number of ordered nonzero summands with sum L :

$$(14) \quad a_t(L) = \frac{1}{q} \sum_{y \in V^{**}} (-1)^{y(L)} \left(\sum_{\ell \neq 0} (-1)^{y(\ell)} \right)^t.$$

The inner sum is N for the trivial character and -1 otherwise. The sum of all nontrivial characters at L is N when $L = 0$ and -1 otherwise. This proves (6), including $t = 0$. Division by the N^t equiprobable histories proves the transition formula. $\square$

Proof of Theorem 1(iii). On the q -point set (7), the probability deviations from $1/q$ are

$$(15) \quad p_t(0) - \frac{1}{q} = \frac{(-1)^t}{qN^{t-1}}, \quad p_t(L) - \frac{1}{q} = -\frac{(-1)^t}{qN^t} \quad (L \neq 0).$$

Half the sum of their absolute values is $1/(qN^{t-1})$, proving (8).

The connected regular crown walk has the uniform stationary law on all $2q$ component vertices. At $t = 1$, one point in the occupied parity half has mass zero and the other N points have mass $1/N$; direct subtraction from $1/(2q)$ gives the first line of (9). For $t \geq 2$, (15) and $N \geq 3$ show that every probability in the occupied half exceeds $1/(2q)$. That half contributes total absolute deviation $1/2$, and the empty half contributes another $1/2$. Dividing by two proves the second line. In particular, the phase limit must not be reported as ordinary mixing. $\square$

4. FULL-CARRIER SPECTRUM AND RECONSTRUCTION

For $S \subseteq E$, define the Boolean character

$$(16) \quad \chi_S(A) = (-1)^{|S \cap A|}, \quad \sigma(S) = \sum_{x \in S} x \in V.$$

These 2^m characters form an orthogonal basis on $\mathbb{F}_2^E$.

Proof of Theorem 1(iv). Convolution by the uniform mask measure makes χ_S an eigenvector with eigenvalue

$$(17) \quad \lambda_S = \frac{1}{N} \sum_{\ell \neq 0} (-1)^{|S \cap H_\ell|}.$$

Modulo two, $|S \cap H_\ell| = |S| + \ell(\sigma(S))$. Character orthogonality therefore gives

$$(18) \quad \lambda_S = \begin{cases} (-1)^{|S|}, & \sigma(S) = 0, \\ (-1)^{|S|+1}/N, & \sigma(S) \neq 0. \end{cases}$$

It remains to count the four cases without hiding a rank assumption. The linear map

$$(19) \quad \mathbb{F}_2^E \longrightarrow \mathbb{F}_2 \oplus V, \quad S \longmapsto (|S| \bmod 2, \sigma(S))$$

has columns $(1, x)$ for $x \in E$. For any $u \neq 0$, choose $x \notin \{0, u\}$; then $(1, x) + (1, x+u) = (0, u)$. These vectors span $\{0\} \oplus V$, and $(1, x) + (0, x) = (1, 0)$ supplies the remaining direction. The map is surjective, so every fibre has size $2^{m-d-1} = K$. Sorting (18) by parity and by whether $\sigma(S)$ vanishes proves (10). The full character basis also proves diagonalizability, so no Jordan block is concealed by the repeated multiplicities. $\square$

Proof of Theorem 1(v). The crown support has degree N because the N nonzero forms give distinct masks. Hence $N + 1 = 2^d$ recovers d . A component has $2(N + 1) = 2q$ states; dividing the supplied total state count by this number recovers K . The argument uses the promised family and makes no inverse claim for an arbitrary graph with the same degree and order. $\square$

5. EXACT CONTROLS, EXCLUSIONS, AND STATUS

The paper-local verifier uses only integer and rational arithmetic and imports no scouting or prior-paper code. It checks 1,095,978 exact assertions: literal carriers, cosets, neighborhoods, and Boolean characters through $d = 4$; ordered histories through five steps in those boxes; and the algebraic formulas through $d = 8$ and sixteen steps. This is falsification pressure, not a proof or an owner-search instrument.

At $d = 1$, the sole projective hyperplane is empty, so the chain is the identity rather than a crown walk. Sampling the zero form for $d \geq 2$ would add an empty mask; toggling hyperplane complements would replace the generators $\mathbf{1} + c_\ell$ by c_ℓ . Neither variant is covered. No claim is made for nonbinary spaces.

The remaining claim is deliberately owner-thin: the literal process plus the disjoint-crown conjugacy, exact labelled history kernel, parity-phase TV, and full-carrier multiplicity lift. Generic Fourier and the finite-group proof shell already appear in the internal P145 line and receive zero separation credit. A bounded literature non-hit is not a novelty result. The author-side gate is OWNER__AMBER, and the external lifecycle is HOLD__EXTERNAL.

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